NodeRag Architecture Evidence

Approved architecture facts used by the Milestone 12 retrieval evaluation

Local service boundary

NodeRag runs as one package with a dedicated package user. PostgreSQL 17 and pgvector are bundled inside the SPK. Qwen3-Embedding-0.6B runs on the NAS CPU and emits 1024-dimensional vectors. The browser interface, REST API, MCP endpoint, worker, embedder, and database share one authorization model rather than private UI-only operations.

Component	Boundary	Release decision
PostgreSQL	Package-private Unix socket	TCP is disabled by default
Embedder	Local framed socket	No runtime model execution off device
Vision	Configured Ollama host	Only selected images may leave the NAS
Gateways	Ports 5300 through 5319	Each listener has a fixed silo set

ARCH-5300 is the exact identifier for the first narrowed gateway port. The control interface uses port 8787 when DSM has not assigned another package port.

Storage and document flow

The DSM-selected package volume is the safe default. Bulk chunks, vectors, indexes, models, and extraction caches may be relocated only after a write, fsync, rename, read, capacity, filesystem, and permission probe succeeds. PGDATA is not placed in a shared folder used by synchronization or media indexing services.

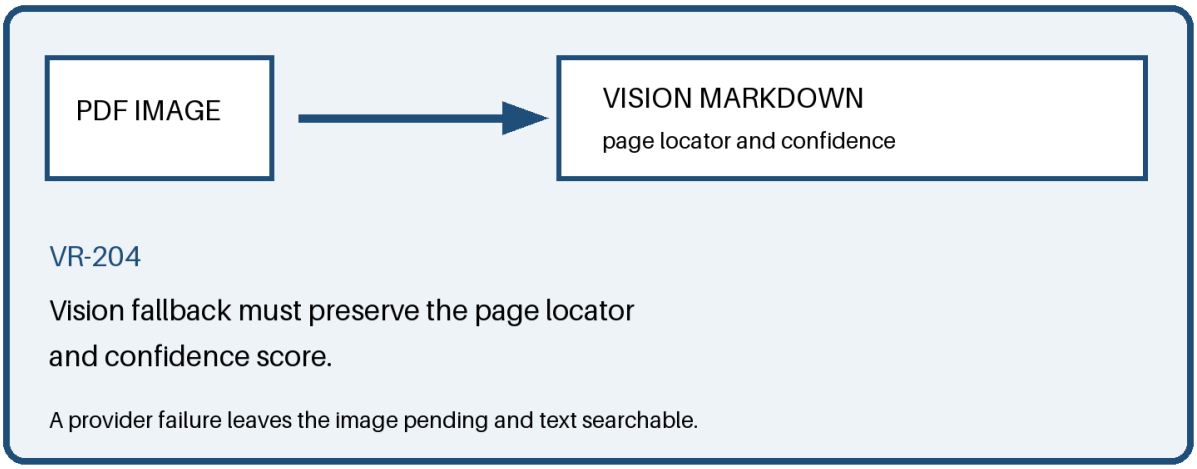
Documents are normalized to NFC before offsets and hashes are computed. PDF and Word pages retain page numbers. Each Excel sheet is treated as an addressable page. Markdown and text use document character offsets when no page exists. A chunk may cross a page boundary, so page_start and page_end are both stored.

Artifact	Integrity rule	Recovery behavior
Original file	SHA-256 must match	Report missing or changed source
Canonical Markdown	Write, fsync, atomic rename	Re-extract from verified original
Embedding cache	Text and model digest key	Recompute only exact cache misses
Vision cache	Image hash and model key	Retry without blocking text ingestion

STOR-0700 identifies the required package-private database socket and data-directory mode. A failed relocation restores the previous path and records an administrator-visible structured error.

Vision recovery evidence

This page contains a raster-only project requirement below. The text inside the image is not part of the PDF text layer. A successful vision path inserts its Markdown at this page locator and records the provider, model, image digest, confidence, and elapsed time.



Decorative images below the configured size threshold are skipped. Repeated images are deduplicated globally by content hash, while cache reference counts ensure that a destructive purge removes vision text that no remaining document references.

Evaluation and performance rules

The golden set contains real questions tied to stable document and location identities. Each run pins the commit, migration head, active build, model manifest, settings revision, corpus snapshot, and search mode. Reports include Recall at 10, mean reciprocal rank, nDCG at 10, zero-result rate, and p50 and p95 latency overall and by the required dimensions.

Decision	Acceptance rule
Cross-silo leakage	Exactly zero
Locator mismatch	Exactly zero
Q4 promotion	At least 1.5 times faster and no more than 0.01 Recall loss
Hybrid promotion	At least 0.02 Recall gain and no more than 2 times p95
Chunk RLS	No more than 15 percent p95 overhead with correct plans

PERF-3000 is the provisional idle search p95 limit in milliseconds. PERF-5000 is the equivalent limit while background ingest is active. These values qualify only the recorded Docker resource envelope and do not become NAS performance promises until final DSM certification.

The nightly run compares the accepted baseline with the current candidate. It alerts on quality, latency, locator, partition coverage, model-version, or cross-silo regressions.